

SDLC Laboratory

Quality Laboratory Manual

Experiment No. 04

To perform the function oriented diagram: DFD and Structured Chart.



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Experiment No. 04

Title of Experiment: To perform the function oriented diagram: DFD and Structured Chart.

Aim of Experiment: To understand and design function-oriented diagrams using Data Flow Diagram (DFD) and Structured diagrams for system analysis and design.

System Requirements – Win 10 and above OS, 4GB RAM, 2.33 GHz Processor

Software/s Requirement – StarUML

Experiment Objectives:

- To understand the importance of DFD and Structured Charts in the software project.
- To understand the components of DFD and Structured Diagram.
- To draw the DFD and Structured diagram for the application to be implemented.

Experiment Outcomes:

- Data Flow Diagram based on the requirement.
- Structured Diagram based on requirement.
- Better and quick understanding of the system.

Theory:

Data Flow Diagram:

A Data Flow Diagram (DFD) is a graphical representation that shows how data moves through a system. It explains how information is input, processed, stored, and output within a system. DFD is mainly used during system analysis and design to understand system functionality clearly.

Components of DFD:

1. External Entity:

An external entity can be a person, organization, or system that is external to the system. From entities, the information flows into the system (known as a source) or the entities receive data from the system (known as a sink). Entities are represented using rectangles

2. Process:

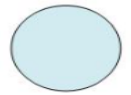
A process is an activity or a function that is carried by the system. Processes manipulate and transform the data. They receive some inputs and generate some outputs.

3. Data Store:

Data stores are used to store the information by the process that can be later retrieved by the same or another process

4. Data Flow:

Data flows describe the flow of information between processes, entities, and data stores. Every data flow name should be a noun. Data flows are represented using arrows.

Notation	De Marco & Yourdon	Gane and Sarson
External Entity		
Process		
Data Store		
Data Flow		

DFD Symbol

Levels of DFD

- It is very difficult to explain all the processes in just one DFD that is why DFDs are expressed in a set of DFD levels. The first step in creating DFDs is to identify the DFD elements (external entities, processes, data stores, and data flows)
- The next steps involve creating the different levels of the DFDs. The highest level DFD depicts the synopsis of the system and with decreasing levels of the DFDs, detailed explanations of each segment of the whole process are described.

1. Level 0 DFD:

These diagrams describe general high-level processes (the first level of numbering is used), external entities, data flows, and data stores. Level-0 DFD is the highest-level view of a system. It shows the entire system as a single process and its interaction with external entities. It provide a basic overview of how data enters and leaves the system without showing internal details.

2. Level 1 DFD

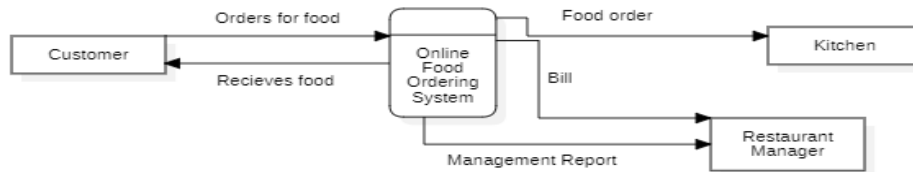
Level – 1 DFD decomposes each parent process of the Level – 0 DFD into more details; into child processes. It also contains data stores, external entities, and data flows. It breaks the main system process into major sub-processes and shows internal working.

3. Level 2 DFD:

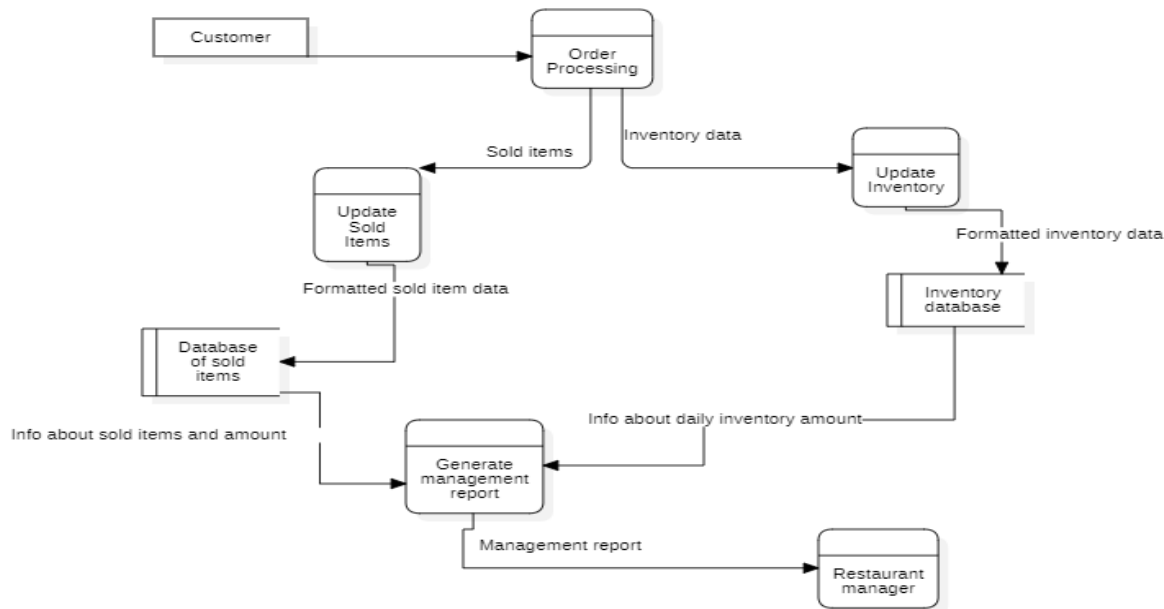
Level-2 DFD is a further detailed breakdown of Level-1 processes. It explains each major process into smaller sub-processes for deeper system understanding. Level 2 show step-by-step internal workflow, decision points, and detailed data movement within a specific process.

An example of different levels of DFDs for Online Food Ordering System

Level-0



Level-1



Task to Perform

Case Study: Online Shopping System

The Online Shopping System is designed to allow customers to browse products, add items to cart and purchase products through an online platform. Traditional shopping requires physical presence which is time-consuming and less convenient. The system aims to provide a secure, user-friendly and efficient online purchasing environment. It also helps sellers manage products, orders and payments digitally. The system improves accessibility, reduces manual work and enhances customer experience.

Task for students

- Perform the function oriented diagram for the above case study by completing the following –

- Problem understanding** – Write a brief problem statement (4-5 lines)
- Identify System Components**-List External entities(minimum 5)

- 3. Draw Data Flow Diagrams (DFD)**-Draw level 0, level 1 and level 2 diagrams
- 4. Develop Structured Chart**-Draw structured chart

Observations:

- System can be represented in different levels of details using DFD levels
- Can start with context level and can be detailed as Level-0, Level-1, Level-2
- Helps to understand the data flows between Entity, Process and Data store.

Conclusion:

The experiment successfully demonstrated, how to draw the different levels of DFD to make the understanding of the system more detailed.

Expected Oral Questions:

1. What is DFD?
2. List out components used in DFD?
3. What is external Entity?
4. What is Process?
5. What is Data Store?
6. What is the difference between Flow chart and DFD?

FAQs in Interview:

1. What is the use of DFD?
2. How DFD is differed from Flowchart?
3. What is the need of different levels of DFD?
4. What is context diagram?